

Solucion-Ejercicio-2.pdf



Anónimo



Ingenieria Estructural



1º Máster en Ingeniería Industrial



Escuela Politécnica Superior. Campus de Leganés
Universidad Carlos III de Madrid



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antes



**Descarga sin publi
con 1 coin**



Después

WUOLAH



Importante

Puedo eliminar la publi de este documento con 1 coin

¿Cómo consigo coins? → Plan Turbo: barato
→ Planes pro: más coins

pierdo
espacio



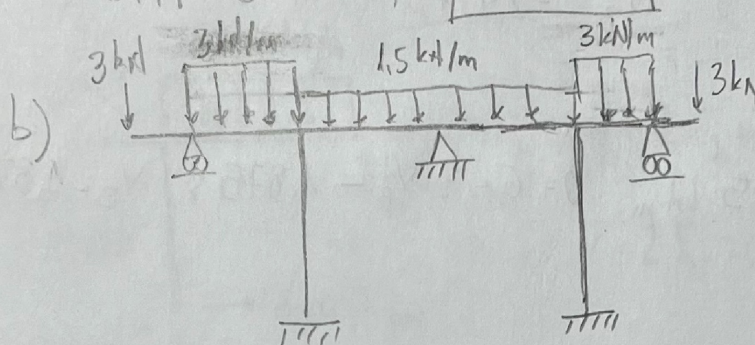
a) $GHE = 10 - 3 = 7$

$GLI = 3(b-1) = 3(8-1) = 21$

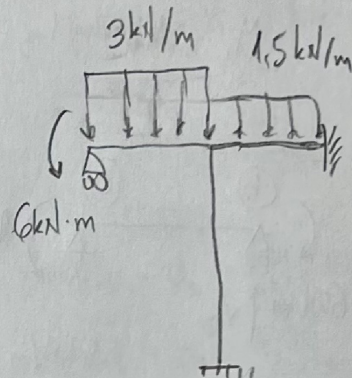
$CI = 3[3(2-1)] + 2[3(3-1)] = 9 + 12 = 21$

$GHI = 0$

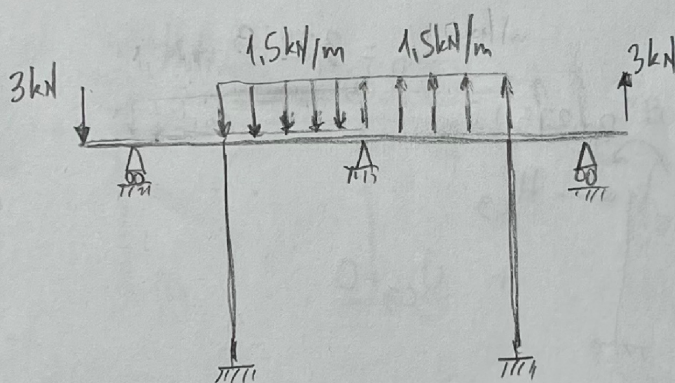
$GH = 7$



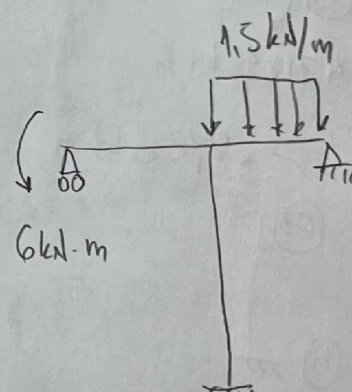
Simétrico



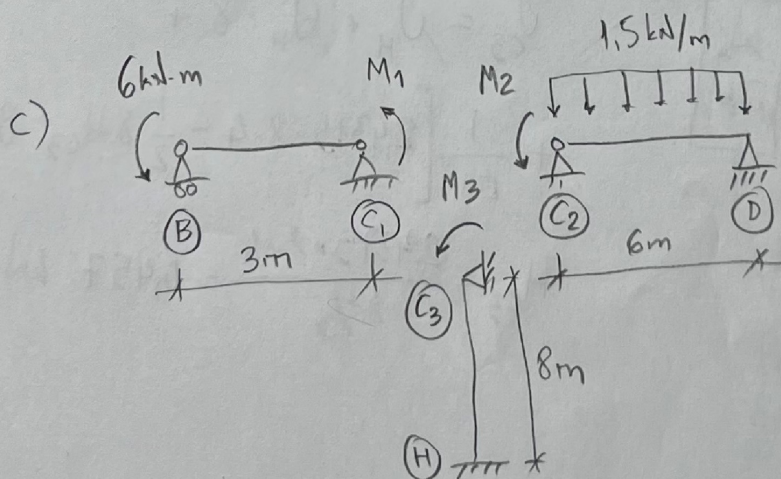
Simplificación simétrico



Antimétrico



Simplificación antimétrico

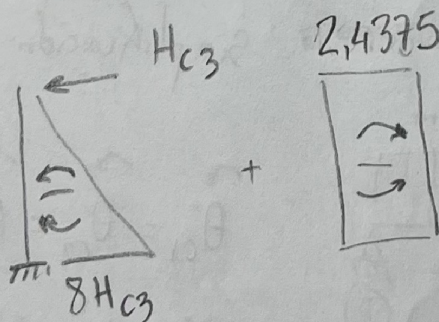
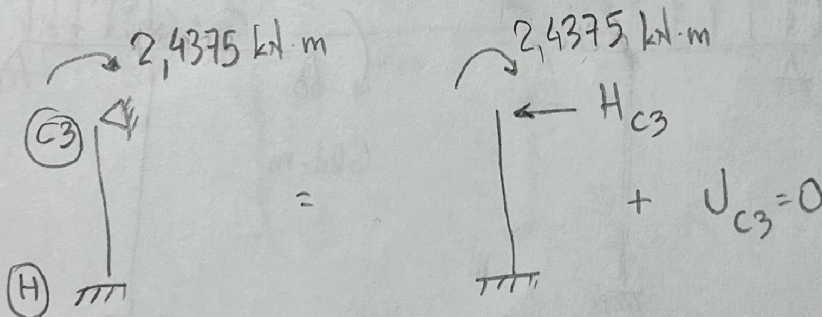
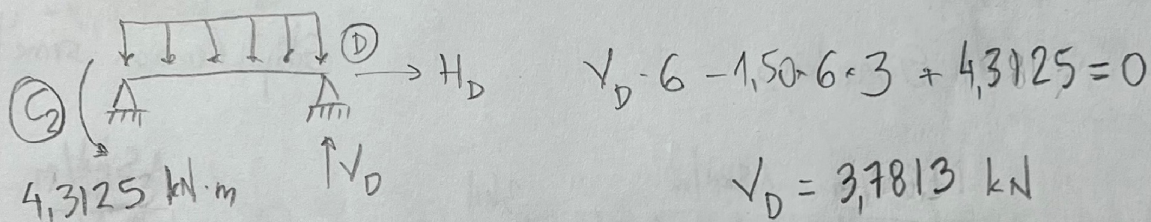
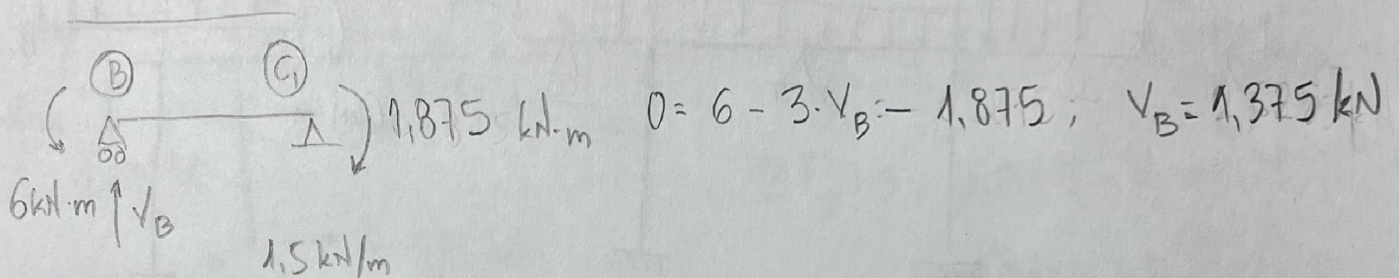


$\theta_{C1} = \theta_{C2} = \theta_{C3}$

$M_1 + M_2 + M_3 = 0$

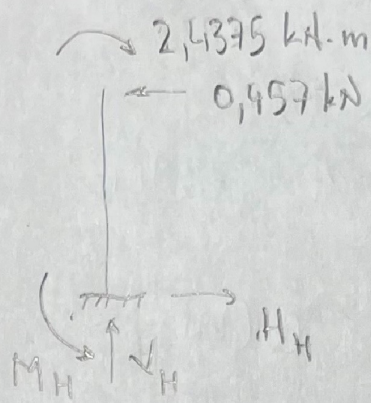
$$\hat{\theta}_{c1} = \frac{-6 \cdot 3}{6EI} + \frac{M_1 \cdot 3}{3EI} ; \hat{\theta}_{c2} = \frac{1,5 \cdot 6^3}{24EI} - \frac{M_2 \cdot 6}{3EI} ; \hat{\theta}_{c3} = + \frac{M_3 \cdot 8}{4EI}$$

$$\left. \begin{array}{l} 3 - M_1 = -2M_3 \\ 13,5 - 2M_2 = -2M_3 \\ M_1 + M_2 + M_3 = 0 \end{array} \right\} \begin{array}{l} M_1 = -1,875 \text{ kN}\cdot\text{m} \\ M_2 = 4,3125 \text{ kN}\cdot\text{m} \\ M_3 = -2,4375 \text{ kN}\cdot\text{m} \end{array}$$



$$U_{C3} = U_H + \cancel{\theta_H \cdot 8} + \frac{1}{EI} \left[2,4375 \cdot 8 \cdot 4 - \frac{1}{2} 8 H_{C3} \cdot 8 \frac{2}{3} 8 \right]$$

$$H_{C3} = \frac{2,4375 \cdot 8 \cdot 4}{8^3/3} = 0,457 \text{ kN}$$

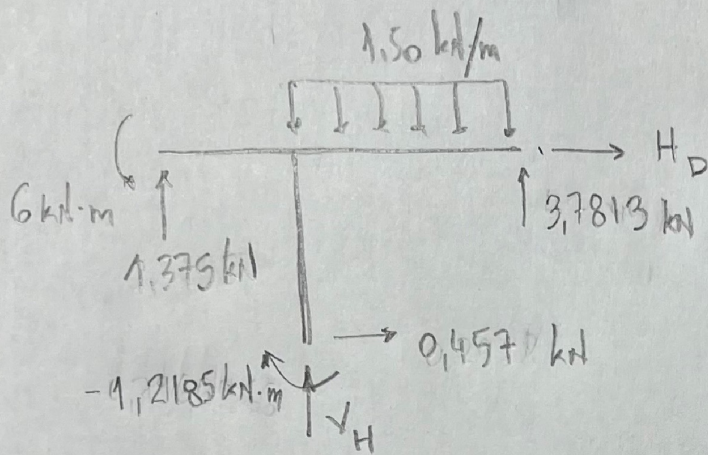


$$\sum M_{c3} = 0 = -2,4375 + 8H_H + M_H$$

$$\sum F_h = 0 = -0,457 + H_H$$

$$H_H = 0,457 \text{ kN} , \quad M_H = 2,4375 - 8 \cdot 0,457$$

$$M_H = -1,2185 \text{ kN}\cdot\text{m}$$

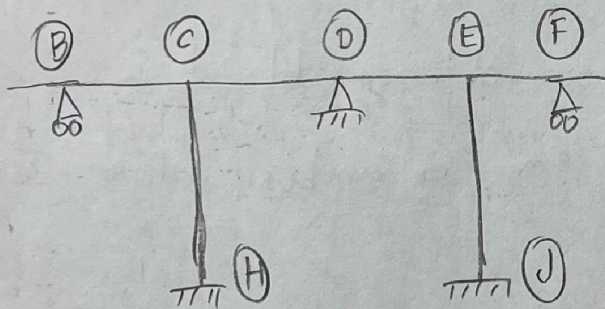


$$\sum F_h = 0 ; \quad H_D = -0,457 \text{ kN}$$

$$\sum F_v = 0 ; \quad 1,375 + V_H + 3,7813 -$$

$$-1,50 \cdot 6 = 0,914$$

$$V_H = 3,8437 \text{ kN}$$



$$V_B = 1,375 \text{ kN}$$

$$V_D = 0 \text{ kN (antimetria)}$$

$$H_D = 2 \cdot (-0,457) = -0,914$$

$$V_F = -1,375 \text{ kN}$$

$$V_H = 3,8437 \text{ kN}$$

$$H_H = 0,457 \text{ kN}$$

$$M_H = -1,2185 \text{ kN}\cdot\text{m}$$

$$V_J = -3,8437 \text{ kN}$$

$$H_J = 0,457 \text{ kN}$$

$$M_J = 1,2185 \text{ kN}\cdot\text{m}$$

Importante

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perdo
espacio



Necesito
concentración

ali ali ooh
esto con 1 coin me
lo quito yo...

WUOLAH

